

MiraTyper



Build your **biological coordinate system**
across *modalities, instruments,*
tissues and diseases

Spatial Data to **Insights...** *Fast*

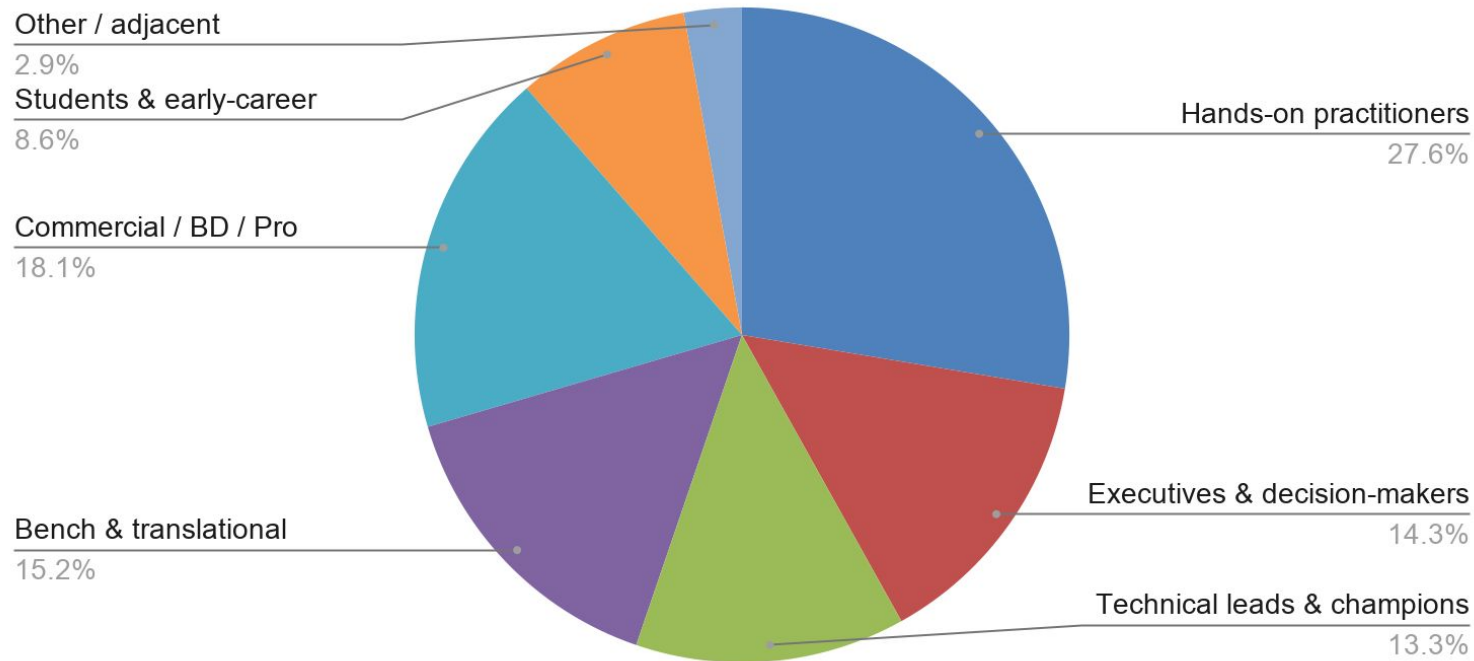
An AI Workshop

RJ Honicky - CTO MiraOmics

AACR: Data download → Results ~1hr



Registered Attendees



Agenda

- Brief introduction to Data / AI / ML for Pharma and MiraTyper
- **Interactive** workshop
 - Git repo - <https://github.com/miraomics/workshop-spatial-ai-2026-06-10>
 - HF dataset - <https://huggingface.co/datasets/honicky/cervical-tls-workshop-crop>
 - Slightly opinionated tool stack - introduced throughout the session
 - Workflow walkthrough - reproduce figures from <https://miraomics.bio/case-studies/cervical-triads-tls/>
- Discussion / Questions / Followups

Some Relevant Quotes

"A phase one trial is minimum budget \$3 million. Once you push something to pre-IND filing, you've got regulatory costs, GXP costs — just that Phase 1 misstep is multi-millions."

"...the thin, skinny layer, that really doesn't give us all that much but gives us an illusion of integration. [MiraTyper] is truly integrating the biology."

"30% of the cells showing cross contamination... 10% of the cells expressing B cell markers where we should be expecting close to 80%... So you don't know if a 0 is actually 0 or a 1 is actually 1, so that's harsh."

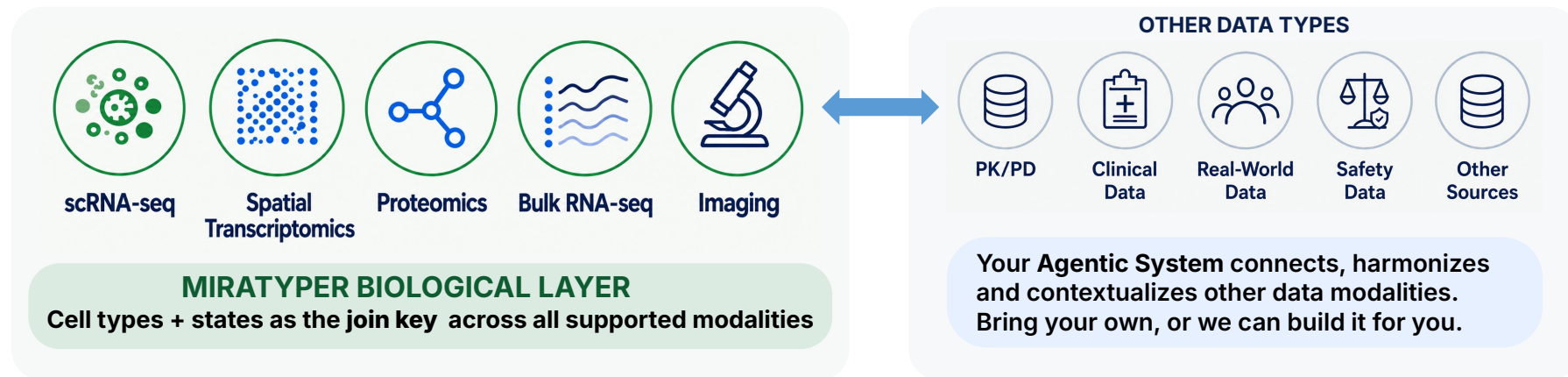
Pharma Data Pain

i.e. why are you here?

- Tying data between clinical and pre-clinical can **improve discovery** and **derisk clinical trials**, but very hard to integrate
- Raw data from spatial instruments are **hard to interpret** - bottlenecked by **overloaded bioinformatics teams**
- **AI is accelerating people's workflows**, but there are too many options, risk of (large) investment without payoff
- ???

Our Differentiation

Integrate Biology at the Data Layer



We natively integrate modalities in the **MiraTyper Biological Layer** to create a unified view of *cell types* and *cell states*.

MIRATYPER APPROACH



- ✓ Integrate at the measurement (data) layer — not the report layer
- ✓ Preserve biology and context across all supported modalities
- ✓ Re-queryable, extensible, and future-proof

Workshop

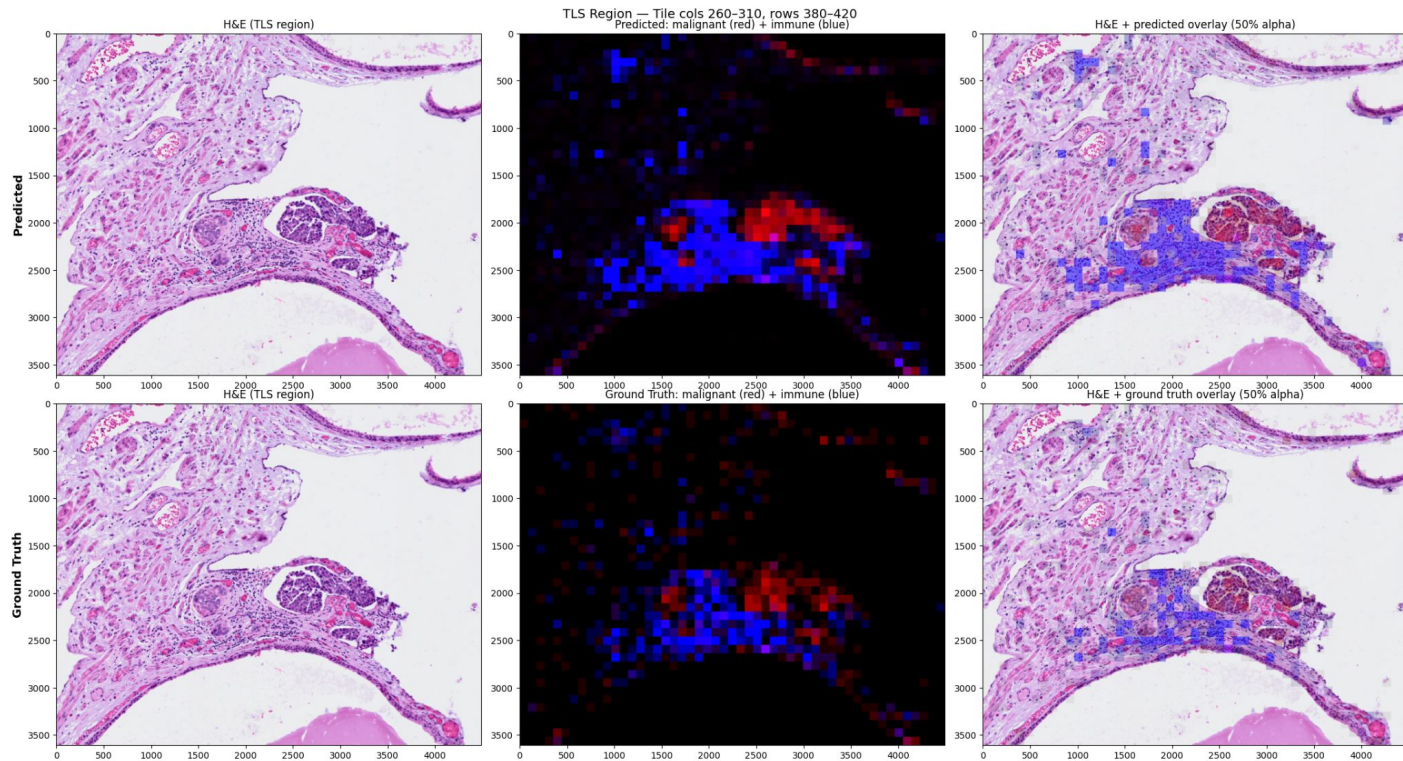
The Biological Coordinate System

Find *common biology* across *modalities* and *technologies*

INTEGRATED DATA LAYER	ENABLES	OUTCOME
 TME composition & state	Patient stratification and response prediction	→ Higher trial success rates, better patients
 Longitudinal integration (pre → clinical → market)	Reverse translation and learning loops	→ Rescue failed assets, improve future design
 Cross-modality cell typing	Consistent biology across assays and teams	→ No re-interpretation across teams, faster alignment
 Cell-type resolved toxicity	Mechanism-level understanding of tox signals	→ Earlier kill decisions, reduced late-stage failures
 Tissue-level & molecular endpoints	Standardized, biologically grounded readouts	→ Faster regulatory alignment, stronger evidence

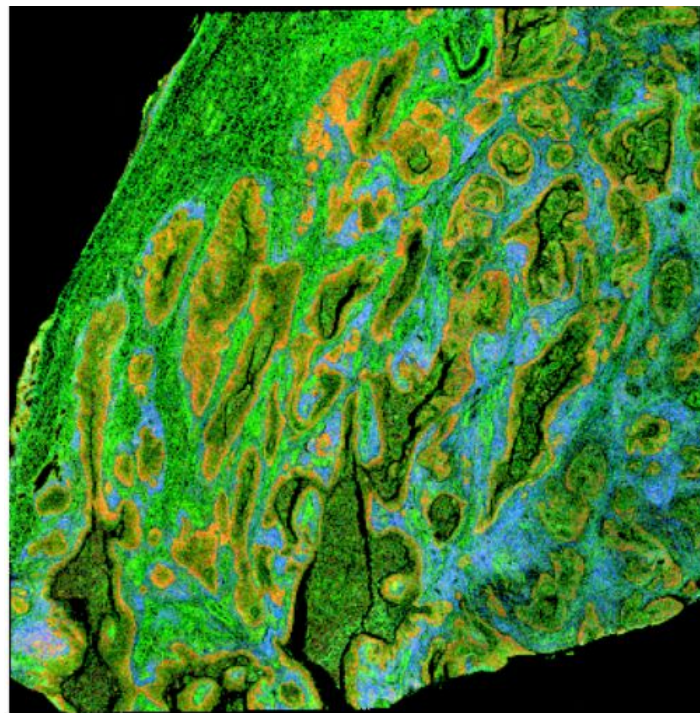
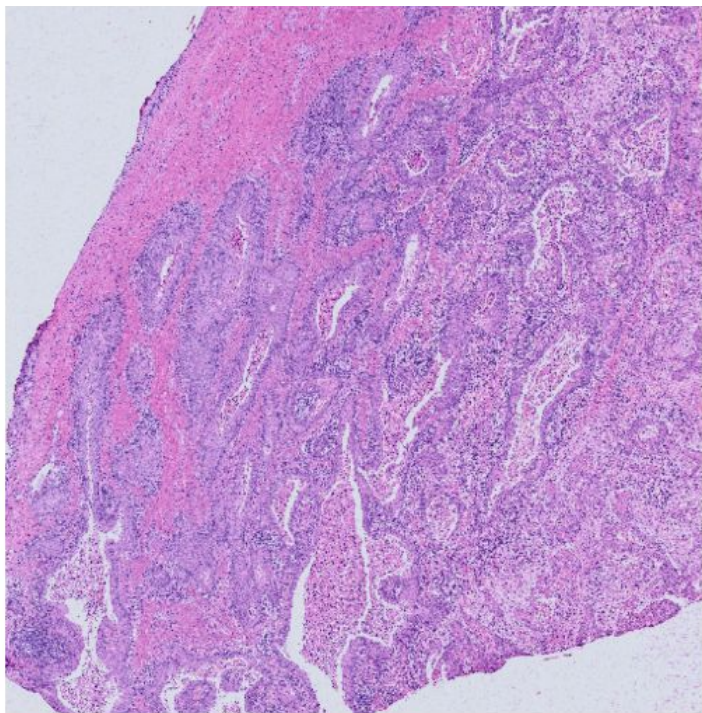
It's the **join key** for cells.

Example: H&E-only *immune* and *malignancy* detection



✓ Deep Ontology ✓ Fine tuning system ✓ Malignant Model

Example: Prediction on Cervical Cancer

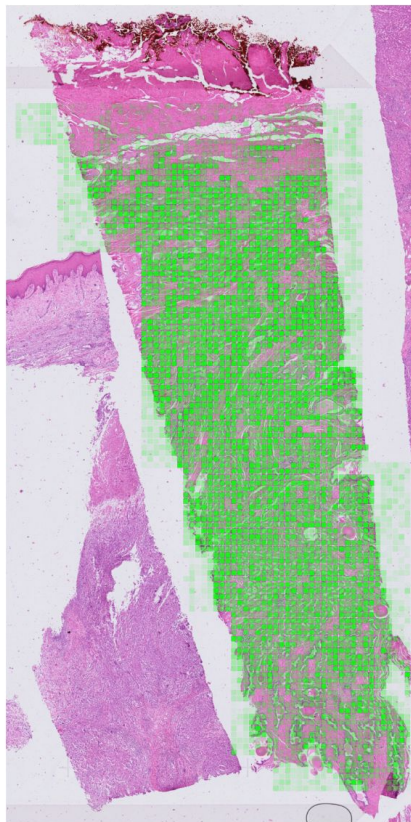


Model validated on Cervical Cancer as well

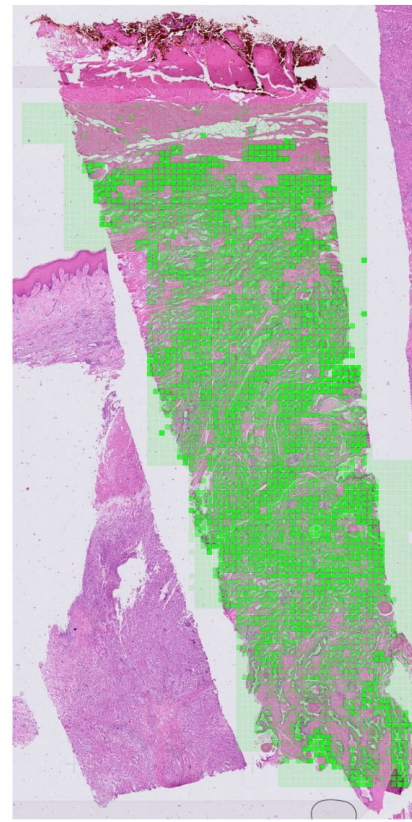
Example: PSMA RNA expression in Prostate Cancer

- Early studies show strong concordance
 - Spearman $\rho = 0.727$ in this example
- Currently evaluating against H&E / IHC serial sections

Predicted PSMA (RNA)



Xenium Reference



Type cells at highest possible resolution

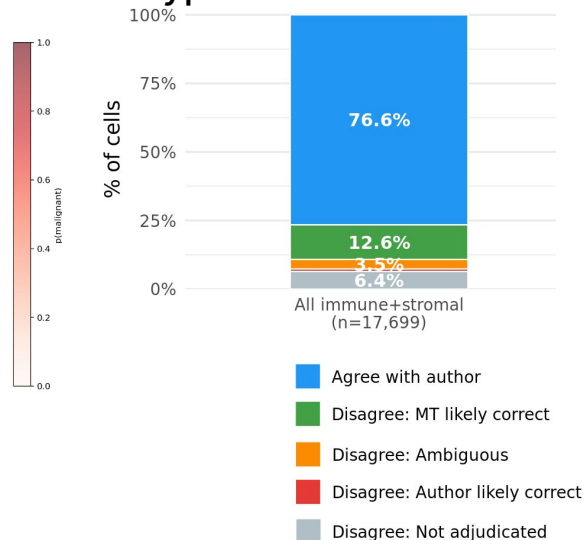
MiraTyper



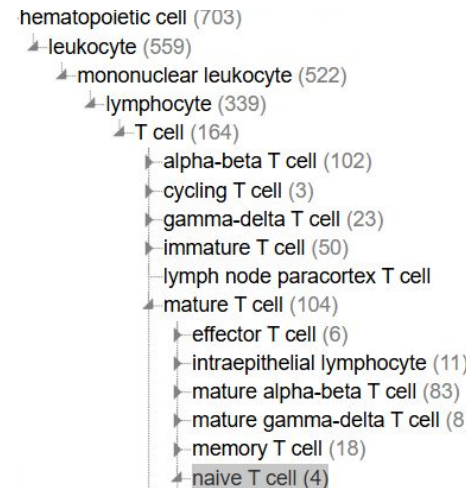
Malignant prediction



Agreement - Author and MiraTyper



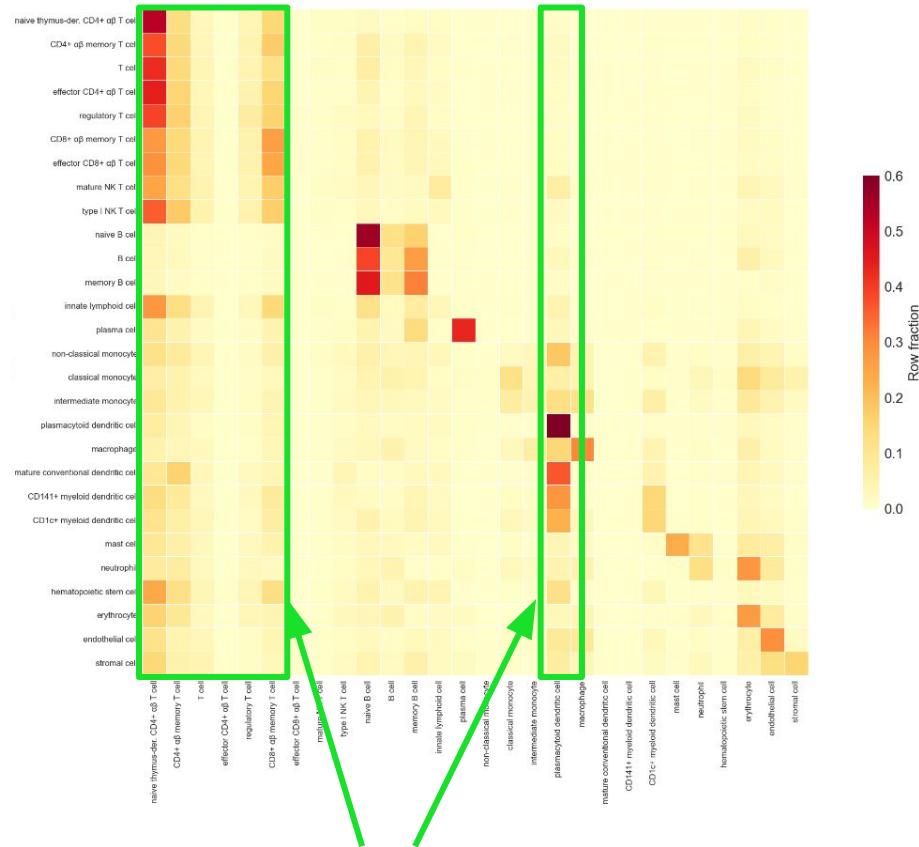
Hierarchy



Typical example: Duarte et al. 2020, Nature Com.
shown >85% accuracy, exceeding author annotation.

Accurate Deep Ontology

- Reference free, accurate (hF1 ~.81), detailed (425 cell types)
- Allows classification of slides in seconds to minutes, insights in minutes to hours
- Allows comparison between instruments (Xenium, CosMX, Atera, Singular) and modalities (scRNA-seq, Spatial, H&E)

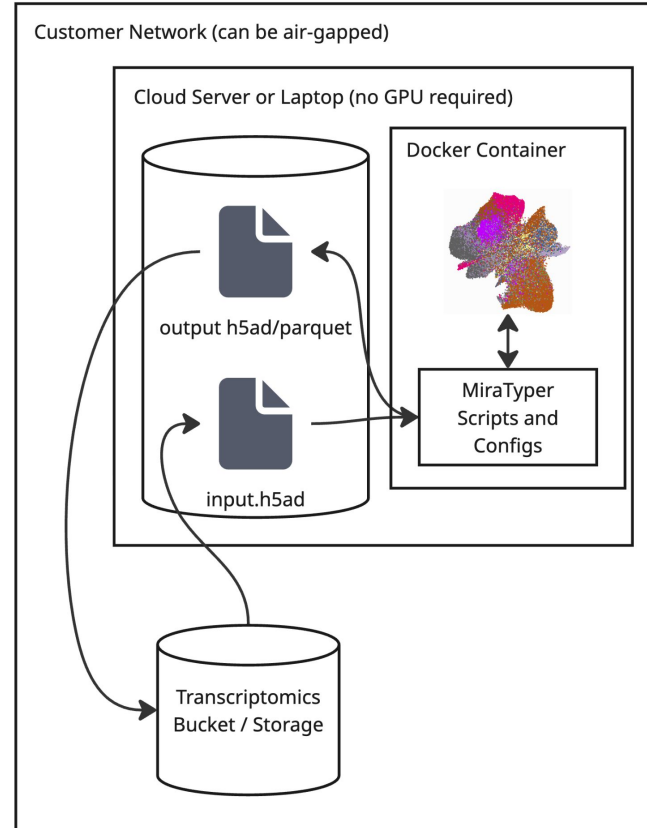


Reference is wrong! <https://miraomics.bio/case-studies/xenium5k-lymph-node/>

Feature details

Fast, local inference

- Turned around Atera slide triad analysis in 1-hour from download!
- Laptop, cloud, embedded are all easy
- Security is easy
- ~100M cells in ~4 hours on moderate hardware @ \$26



MiraTyp^{er}



Interested in turning you **data** into **insights**?

[Set up a call](#)

info@miraomics.bio